

# Insect Species Classification Using MFCC and Log-Mel Features: A Comparative Study on InsectSet66

Ali Rastegar Mojarad  
Computer Science Department  
University of Milan  
Milan, Italy  
ali.rastegar@studenti.unimi.it

**Abstract**—This project investigates acoustic insect species classification on InsectSet66, a benchmark of 66 Orthoptera and Cicadidae species. We extract Mel-frequency cepstral coefficients (MFCC) and log-mel spectrograms from 5-second audio segments and evaluate two complementary approaches. First, we replicate the mel-spectrogram CNN baseline of Faiss et al. [1], achieving 60.4% test accuracy. Second, we apply k-means clustering, Support Vector Machines (SVM), and Random Forest (RF) on MFCC statistical features. SVM achieves 76.7% test accuracy and 0.71 macro-F1, outperforming the unaugmented CNN. RF feature importance provides interpretable insight into which cepstral coefficients drive species separation.

**Index Terms**—Insect sound classification, bioacoustics, MFCC, log-mel spectrogram, InsectSet66, SVM, Random Forest

## I. INTRODUCTION

Insects play a critical role in ecosystems as pollinators, decomposers, and indicators of biodiversity. However, insect populations are declining globally, and monitoring them at scale is difficult. Passive acoustic recording is a practical approach: many insect species produce species-specific sounds that can be captured in the field. Automated recognition of these sounds enables large-scale, cost-effective biodiversity monitoring.

The task of classifying insect species from audio is challenging. Species within the same genus produce acoustically similar calls, recordings vary in quality, and datasets are often imbalanced. InsectSet66 [2] provides a curated benchmark of 66 Orthoptera and Cicadidae species with official train, validation, and test splits.

Faiß and Stowell [1] proposed a convolutional neural network (CNN) trained on log-mel spectrograms as the reference system for this dataset. Their 5-layer mel-CNN achieves 82% test accuracy with data augmentation (colored noise and impulse responses). Classical machine learning methods, while generally less accurate than deep models, offer interpretability and are useful for understanding which acoustic features drive classification.

In this project we: (1) replicate the log-mel CNN baseline from Faiß and Stowell [1]; (2) extract MFCC features and apply k-means clustering to visualize the acoustic feature space; (3) train SVM and Random Forest classifiers on MFCC statistics; (4) analyze model interpretability through RF feature importance and confusion patterns.

## II. DATASET

We use InsectSet66 [2], a public dataset of field recordings from 66 insect species belonging to the orders Orthoptera (grasshoppers, crickets) and Cicadidae (cicadas). The dataset contains 2,887 mono WAV files with official train, validation, and test splits. Table I shows the number of recordings per subset. Species representation is imbalanced, ranging from 10 to 160 recordings per species (median 35). This motivates reporting macro-averaged F1 alongside accuracy.

TABLE I  
INSECTSET66 RECORDING COUNTS BY SUBSET.

| Subset     | Recordings | Chunks |
|------------|------------|--------|
| Train      | 1,752      | 41,312 |
| Validation | 579        | 12,587 |
| Test       | 556        | 10,391 |
| Total      | 2,887      | 64,290 |

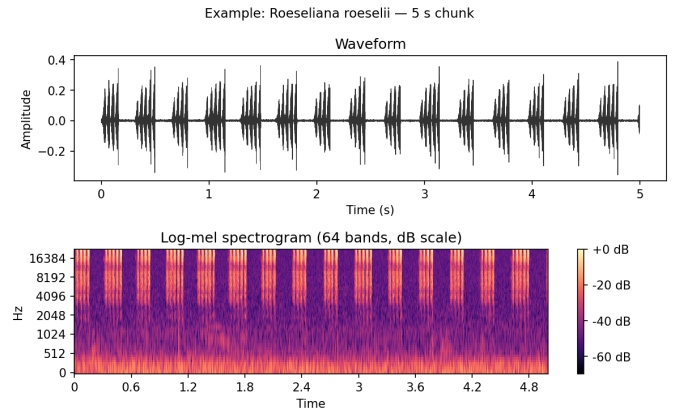


Fig. 1. Example log-mel spectrogram (64 mel bands, log-power in dB). Time is on the horizontal axis and mel frequency on the vertical axis.

## III. METHODS

### A. Segmentation

Each recording is split into 5-second chunks with 3.75 s overlap (1.25 s advance step). Short recordings are loop-padded to reach 5 s. The final partial segment is kept if it exceeds 1.25 s. This yields 64,290 chunks across all subsets, as shown in Table I. All waveforms are resampled to 44.1 kHz mono. The official file-level splits are preserved so that chunks from the same recording always belong to the same subset.

### B. Feature extraction

1) *Log-mel spectrogram (CNN baseline)*: Each 5-second chunk is transformed into a log-mel spectrogram. A short-time Fourier transform (STFT) is applied with  $n_{\text{FFT}} = 1000$ , hop length 147, and window length 294. The power spectrum is mapped to 64 mel-frequency bands and converted to a decibel scale using `AmplitudeToDB` with `top_db=80`. The resulting  $64 \times 1501$  image is the input to the CNN. Figure 1 illustrates an example log-mel spectrogram.

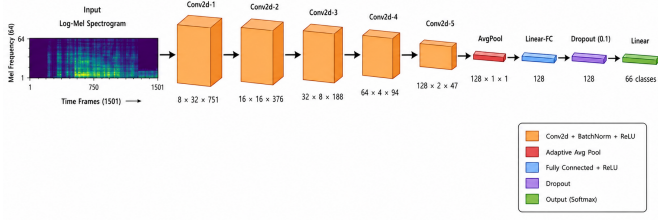


Fig. 2. Proposed CNN architecture for log-mel spectrogram features. The input  $1 \times 64 \times 1501$  tensor passes through five Conv2d blocks (each followed by BatchNorm and ReLU), with stride 2 for progressive downsampling. Global average pooling collapses the spatial dimensions to a 128-d vector, which is passed through dropout and a linear layer to produce 66 class logits.

2) *MFCC statistical features (classical pipeline)*: We extract 20 Mel-frequency cepstral coefficients (MFCCs) per frame using  $n_{\text{FFT}} = 2048$  and hop length 512 [3]. First- and second-order delta coefficients capture temporal dynamics. For each chunk, we compute the mean and standard deviation of each coefficient across all frames, yielding a fixed 120-dimensional feature vector: 20 coefficients  $\times$  3 groups (MFCC,  $\Delta$ ,  $\Delta^2$ )  $\times$  2 statistics (mean, std). Feature vectors are standardized to zero mean and unit variance before SVM training.

#### C. Models

1) *Log-mel CNN (baseline replication)*: We replicate the 5-layer convolutional classifier from Faiss et al. [4]. The network applies five Conv2d blocks (8, 16, 32, 64, 128 channels) with BatchNorm and ReLU activations, followed by global average pooling and a linear output layer for 66 classes. Training uses Adam with OneCycleLR scheduling and early stopping on validation loss (patience 8). Data augmentation (colored noise and impulse responses) is disabled in our replication due to missing assets. Figure 2 illustrates the full network architecture.

2) *Support Vector Machine*: We train an SVM with an RBF kernel,  $C = 10$ , and `class_weight=balanced` on the scaled 120-dimensional MFCC vectors. The balanced class weighting compensates for species imbalance.

3) *Random Forest*: We train a Random Forest with 200 trees and `class_weight=balanced` on the unscaled MFCC vectors. Feature importance scores (Gini impurity) are used for interpretability analysis.

#### D. Evaluation metrics

We report **accuracy** and **macro-averaged F1** on the held-out test set. Macro-F1 averages F1 equally across all 66 classes and is more informative than accuracy under class imbalance. Confusion matrices are visualized as heatmaps.

#### IV. UNSUPERVISED ANALYSIS

We apply PCA to reduce both feature representations to 2D and run k-means clustering on the training set. We use  $k = 2$  for visualization (corresponding to the two insect orders) and  $k = 66$  to compute Normalized Mutual Information (NMI) against true species labels. NMI ranges from 0 (no alignment) to 1 (perfect alignment).

For MFCC features, the first two principal components explain 31.1% of variance and  $\text{NMI} = 0.58$ . For log-mel features,  $\text{NMI} = 0.587$ , marginally higher than MFCC. Both feature types show partial order-level separation: Cicadidae form a compact cluster while

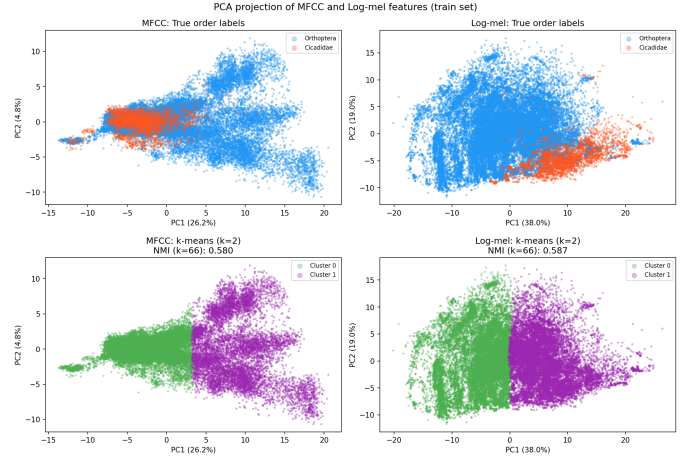


Fig. 3. PCA projection of MFCC (left) and log-mel (right) features on the training set. Top row: true order labels. Bottom row: k-means ( $k = 2$ ) cluster assignments. NMI scores ( $k = 66$ ) are shown in panel titles.

TABLE II  
CLASSIFICATION PERFORMANCE ON INSECTSET66 (66 CLASSES). BOLD INDICATES BEST RESULT PER COLUMN AMONG OUR MODELS.

| Model                | Features       | Validation  |              | Test        |             |
|----------------------|----------------|-------------|--------------|-------------|-------------|
|                      |                | Acc         | Macro-F1     | Acc         | Macro-F1    |
| SVM (RBF)            | MFCC           | 0.744       | <b>0.633</b> | 0.767       | 0.711       |
| Random Forest        | MFCC           | 0.748       | 0.601        | 0.754       | 0.637       |
| Mel-CNN (replicated) | Log-mel        | —           | —            | 0.604       | 0.514       |
| Mel-CNN [1]          | Log-mel + aug. | <b>0.81</b> | —            | <b>0.82</b> | <b>0.74</b> |

Orthoptera spread more widely. The high NMI confirms that both feature types encode substantial species-discriminative information without supervision.

#### V. CLASSIFICATION RESULTS

Table II summarizes validation and test performance for all models. SVM achieves the highest test accuracy (76.7%) and macro-F1 (0.71). Random Forest reaches 75.4% accuracy and 0.64 macro-F1. Both classical models outperform our replicated mel-CNN (60.4%). The CNN was trained without data augmentation; Faiss and Stowell [1] report 82% (mel-5, with augmentation, median of 3 seeds).

Figures 4 and 5 show the test-set confusion matrices. The diagonal is dominant, indicating most chunks are correctly classified. Errors are concentrated among a few acoustically similar species, consistent with the dataset’s class imbalance.

#### VI. MODEL INTERPRETABILITY

Random Forest provides Gini feature importance scores that indicate which MFCC statistics contribute most to classification. Figure 6 shows the top 20 features. The most important features are low-order MFCC means (coefficients 0–7), which capture the coarse spectral envelope of insect calls. Delta coefficients (`delta1_std`, `delta2_std`) also appear in the top 20, indicating that temporal dynamics contribute to species discrimination.

This analysis is not possible with the CNN baseline, which operates on raw spectrogram images without explicit feature-level attribution. The RF model thus provides an interpretable complement to the deep learning approach.

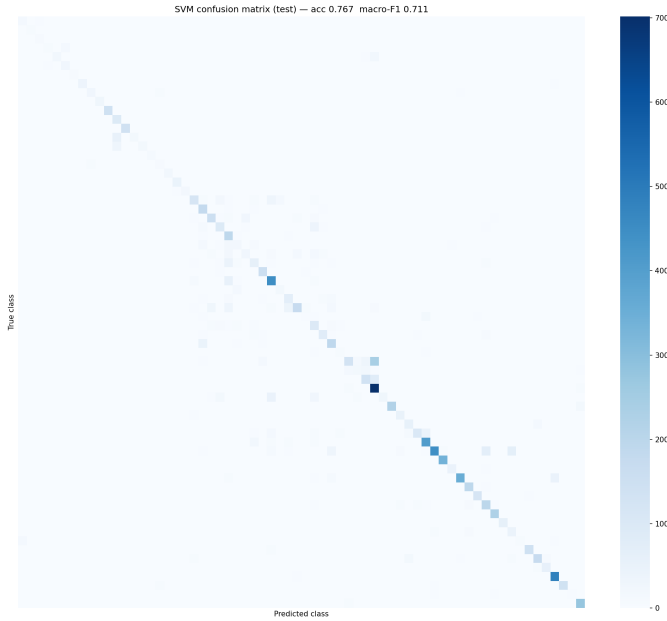


Fig. 4. SVM confusion matrix on the test set (66 classes). Accuracy: 76.7%, macro-F1: 0.71.

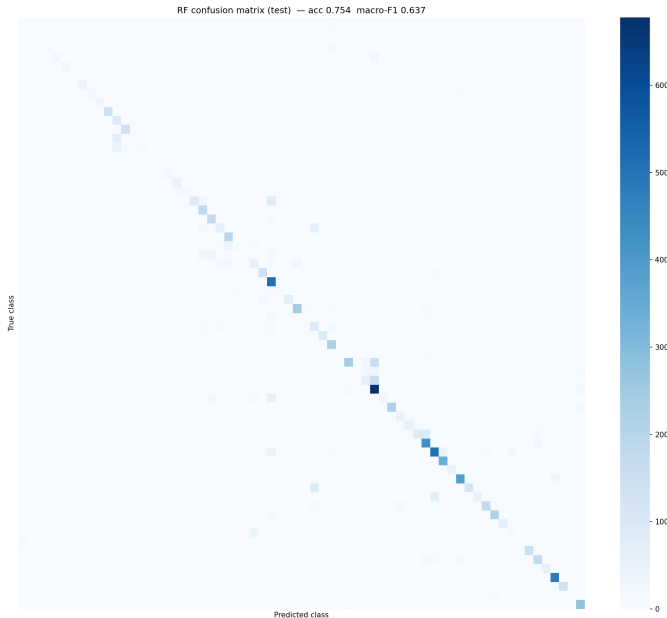


Fig. 5. Random Forest confusion matrix on the test set (66 classes). Accuracy: 75.4%, macro-F1: 0.64.

## VII. DISCUSSION

**Classical vs. deep learning.** SVM and RF on MFCC features outperform our replicated CNN (76.7% vs. 60.4%). This does not imply that classical methods are generally better than deep learning for audio. The CNN was trained without data augmentation (colored noise and impulse responses), which Faiß and Stowell [1] report as essential for achieving 82%. With augmentation, the CNN would likely outperform the classical models.

**Replication gap.** Our replicated CNN achieves 60.4% compared to 82% in Faiß and Stowell [1]. The primary cause is the absence of

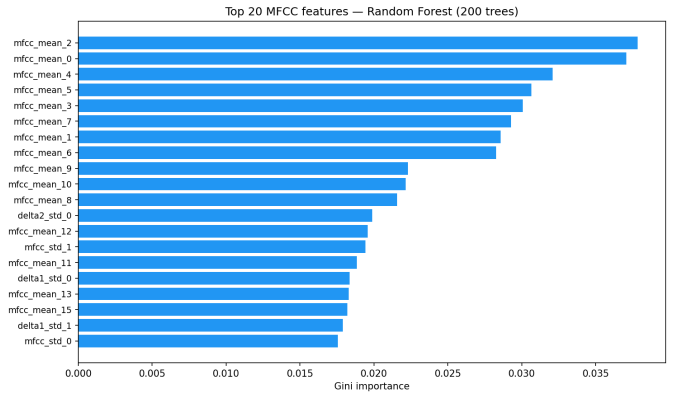


Fig. 6. Top 20 MFCC features by Gini importance (Random Forest, 200 trees). Low-order MFCC means dominate, reflecting their importance for encoding spectral shape.

augmentation. Secondary factors include training on a single random seed and slight differences in hardware (Apple MPS vs. CUDA). We document this limitation rather than attempting to close the gap, as it falls outside the scope of this project.

**MFCC feature space.** The NMI of 0.58 shows that MFCC statistics encode substantial species-discriminative information. Low-order MFCC means, which capture the coarse spectral envelope, are the most informative features. This aligns with the acoustic properties of insect calls, where species are distinguished primarily by call frequency, chirp rate, and syllable structure.

**Class imbalance.** The dataset is imbalanced (10–160 recordings per species). We use `class_weight=balanced` for both SVM and RF to compensate. Macro-F1 is a more reliable metric than accuracy in this setting because it weights each class equally.

**Interpretability trade-off.** RF provides explicit feature importance while the CNN does not. In ecological applications, understanding which acoustic cues drive a prediction is valuable for validating the model and building trust with domain experts. The classical pipeline is therefore a practical complement to the deep learning baseline, not just a weaker alternative.

## VIII. CONCLUSION

We evaluated two approaches for 66-class insect species classification on InsectSet66. The replicated log-mel CNN achieves 60.4% test accuracy without data augmentation. SVM and Random Forest on MFCC statistical features achieve 76.7% and 75.4%, respectively, outperforming the unaugmented CNN while remaining fully interpretable. K-means clustering with NMI = 0.58 confirms that MFCC features encode species-discriminative acoustic structure even without supervision. Random Forest feature importance shows that low-order MFCC means, which capture spectral envelope shape, are the most informative cues for insect species identification.

Future work could close the gap to the published 82% result [1] by enabling data augmentation, or explore transformer-based models such as PaSST, which have shown strong performance on related bioacoustic datasets.

## REFERENCES

- [1] M. Faiß and D. Stowell, “Adaptive representations of sound for automatic insect recognition,” *PLOS Computational Biology*, vol. 19, no. 10, p. e1011541, 2023.
- [2] M. Faiss, “Insectset47 & insectset66: Expanded datasets for automatic acoustic identification of insects,” 2023. [Online]. Available: <https://zenodo.org/records/8252141>

- [3] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar, E. Battenberg, and O. Nieto, “librosa: Audio and music signal analysis in python,” in *SciPy*, 2015, pp. 18–24.
- [4] M. Faiss, “Insectset47-insectset66-adaptive-representations-of-sound-for-automatic-insect-recognition,” <https://github.com/mariusfaiss/InsectSet47-InsectSet66-Adaptive-Representations-of-Sound-for-Automatic-Insect-Recognition>, 2023.